

# EEBus UC Implementation Guideline

## Limitation of Power Consumption

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## 1 Scope of the document

This document is an implementation guideline for the Use Case "Limitation of Power Consumption" (short-name for this document: IG-LPC). It details how to interpret specific concepts and data points and gives examples and best practices on topics reported as probably difficult to understand or where misinterpretations could occur. It also puts the use case in the context of country-specific regulations.

The sections of this document clarify the [LPC] Use Case specification. They are, together with [LPC], the basis for testing procedures of EEBUS devices (e.g., in the Living Lab Cologne (LLC)).

Chapter 2 focuses on clarifications on the high-level part of the Use Case [LPC].

Chapter 3 focuses on clarifications on the technical part of the Use Case [LPC].

Chapter 4 defines country specific rules.

This document will be continuously updated as new issues requiring clarification are identified.

### 1.1 References

#### 1.1.1 EEBUS documents

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| [ProtocolSpecification] | EEBus_SPINE_TS_ProtocolSpecification.pdf (V1.2.0)     |
| [ResourceSpecification] | EEBus_SPINE_TS_ResourceSpecification.pdf (V1.2.0)     |
| [SHIP]                  | EEBus_SHIP_TS_Specification_v1.0.1.pdf                |
| [LPC]                   | EEBus_UC_TS_LimitationOfPowerConsumption.pdf (V1.0.0) |
| [LPP]                   | EEBus_UC_TS_LimitationOfPowerProduction.pdf (V1.0.0)  |

#### 1.1.2 Normative references

|                   |                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| [RFC2119]         | IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels. Please see section 1.3.1 for details.          |
| [BK6-22-300]      | Addition to German law EnWG § 14a defining requirements for building's devices to be limitable by the DSO (27 November 2023). |
| [VDE-AR-E 2829-6] | The VDE-AR-E 2829-6 standardizes the communication interface for data exchange between customer systems and grid operators.   |

### 1.2 Terms and definitions

#### ACK

Abbreviation for "Acknowledgement"

**Actor**

An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

**BNetzA**

Abbreviation for "Bundesnetzagentur (für Elektrizität, Gas, Telekommunikation, Post und Eisenbahnen)" (Federal Network Agency for Electricity, Gas, Telecommunications, Post and Railway), a German authority for all kinds of public grids and networks.

**CEM**

Abbreviation for "Customer Energy Manager". The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimised operation of the connected devices by harmonising energy demand and availability.

**Control box**

A control box can be part of a smart meter or of a smart meter gateway or a separate device. This depends on the country specific regulations, the local setup, etc.

**Controllable System**

Actor name for a logical or physical appliance or system that is remote controllable (e.g. via power limits).

**CS**

Abbreviation for "Controllable System"

**EG**

Abbreviation for "Energy Guard"

**Energy Guard**

Actor name for a logical or physical unit capable of managing the energy demand of connected devices.

**EnWG**

Abbreviation for the German law "Energiewirtschaftsgesetz" (Energy Industry Act)

**EVSE**

Abbreviation for "Electric Vehicle Supply Equipment"

**DSO**

Abbreviation for "Distribution System Operator"

**EMS**

Abbreviation for "Energy Management System"

**GCP**

Abbreviation for "Grid Connection Point"

**IG-LPC**

Abbreviation for "Implementation Guideline for Use Case Limitation of Power Consumption" (short name of this document)

**108 LLC**

109 Abbreviation for "Living Lab Cologne" (see <https://www.livinglabcologne.com> for further  
110 information)

**111 LPC**

112 Abbreviation for the Use Case "Limitation of Power Consumption"

**113 LPP**

114 Abbreviation for the Use Case "Limitation of Power Production"

**115 MGCP**

116 Abbreviation for the Use Case "Monitoring of Grid Connection Point"

**117 MPC**

118 Abbreviation for the Use Case "Monitoring of Power Consumption"

**119 NACK**

120 Abbreviation for "No Acknowledgement"

**121 Scenario**

122 Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more  
123 quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or  
124 optional.

**125 SPINE**

126 Abbreviation for "Smart Premises Interoperable Neutral-message Exchange": Technical Specification  
127 of EEBus Initiative e.V.

**128 UC**

129 Abbreviation for "Use Case"

130

**131 1.3 Requirements****132 1.3.1 Requirements wording**

133 The following keywords are used:

- 134 - SHALL
- 135 - SHALL NOT
- 136 - SHOULD
- 137 - SHOULD NOT
- 138 - MAY

139 Note: They apply only if written in capital letters.

140 For the meaning of the keywords, please refer to [RFC2119].

141

**1.3.2 Notes on requirements in this document**

This document clarifies requirements of [LPC], SPINE ([ProtocolSpecification], [ResourceSpecification]), or [SHIP]. Regarding these specifications, the keywords "SHALL" and "SHALL NOT" are already covered by mandatory requirements of these specifications.

Regarding other specifications (e.g. [BK6-22-300]), the keywords "SHALL" and "SHALL NOT" express mandatory requirements taken from these specifications or considered essential by the EEBUS Initiative e.V.

## 2 Clarifications on the high-level part of the Use Case LPC

### 2.1 Intended purpose of the Use Case LPC

The Use Case [LPC] SHALL only be used for purposes specified in [LPC] (stabilization of the public electricity grid etc.; see section 2.1 in [LPC]). It SHALL NOT be used for other means like self-consumption optimisation, cost savings, etc. (for this, other Use Cases are available).

### 2.2 Duration of a limit

An Energy Guard SHALL NOT send an Active Power Consumption Limit with a duration set to "0".

The following rules clarify in detail situations with duration set to "0":

- 1) If Actor Controllable System receives an Active Power Consumption Limit with a duration set to "0", the limit SHALL immediately be deactivated by the Controllable System (see [LPC-007]). If the received write command contains a duration set to "0" AND also a request to set the Active Power Consumption Limit to "activated", the Actor Controllable System SHALL ignore the request to set it to "activated" and deactivate the limit instead.
- 2) An Energy Guard is only permitted to send an Active Power Consumption Limit with a duration set to "0" if the limit is set to "deactivated" with the same command.
- 3) If an Energy Guard wants to deactivate an Active Power Consumption Limit, the limit can be deactivated explicitly with [LPC-008].

Note: If Actor Energy Guard does not want the Active Power Consumption Limit to depend on a duration value anymore, it needs to remove/unset the duration<sup>\*1</sup>. Setting a duration to "0" does NOT remove/unset the duration!

<sup>\*1</sup>: How the removal of a duration is done depends on the technical realisation of the Use Case. In case of SPINE, the according procedure is described in section 3.4.1.4 of [LPC].

### 2.3 Heartbeat and timeout to detect missing Active Power Consumption Limit

The Use Case [LPC] specifies situations where Actor Controllable System receives a heartbeat but no following Active Power Consumption Limit (within a given duration). The following excerpt from [LPC] gives an example for such a situation:

*"The Controllable System MAY leave the "failsafe state" and switch into "unlimited/autonomous" state if the Energy Guard's Heartbeat is received again, but no write command on the Active Power Consumption Limit is received within 120s ([LPC-921])."*

(in section 2.3 of [LPC] this is enumerated as transition 10).

This rule means that every heartbeat that is received by Actor Controllable System in "failsafe state" virtually instantiates its own timer to check if also an Active Power Consumption Limit is received latest 120s after the corresponding heartbeat was received or not.

This means that already the first heartbeat received in "failsafe state" permits Actor Controllable System to switch into state "unlimited/autonomous" if no Active Power Consumption Limit is received within 120s since this heartbeat.

It DOES NOT mean that a second heartbeat virtually disables the timer that belongs to the first heartbeat.

Note: The above-mentioned excerpt does not force Actor Controllable System to switch to state "unlimited/autonomous" if the condition is fulfilled. In this particular case, the Actor may well remain in "failsafe state" and may switch to state "unlimited/autonomous" later, unless another event occurs where rules of [LPC] require a specified action (e.g. the Actor Controllable System decides to remain in "failsafe state", but later a heartbeat and power limit within 120s are received which require a switch to state "limited").

Of course, the explanations given above do not only apply to "transition 10", but to comparable situations as well.

## **2.4 ACK/NACK behaviour of an EMS as Controllable System**

This section provides recommendations for an EMS as Actor Controllable System to report reliable ACK/NACK messages towards the grid's Actor Energy Guard.

The Use Case [LPC] clearly limits possibilities to reject (NACK) commands from an Energy Guard. However, it is also not valid for an EMS to accept (ACK) commands unconditionally. The following recommendations comply with [LPC] rules and provide a reliable ACK/NACK-based concept.

Note: A Controllable System sends an ACK message only if it will follow the limit. Otherwise, a NACK is sent. Hence, an ACK is NOT only an acknowledgement of receipt!

An example for the importance of the ACK concept is given in section 4.1.5.

As soon as an EMS receives a new activated power limitation, it has to evaluate whether the overall limit can be kept by the according devices\*<sup>1</sup> or not. The best way an EMS can ensure this is by calculating the limits for each device, sending the limits to the devices and analysing their responses (ACK = limit accepted or NACK = limit rejected). Then, the EMS can send a corresponding response to the grid's Actor Energy Guard (ACK, if the overall limit can be kept\*<sup>2</sup> or NACK, if the overall limit cannot be kept).

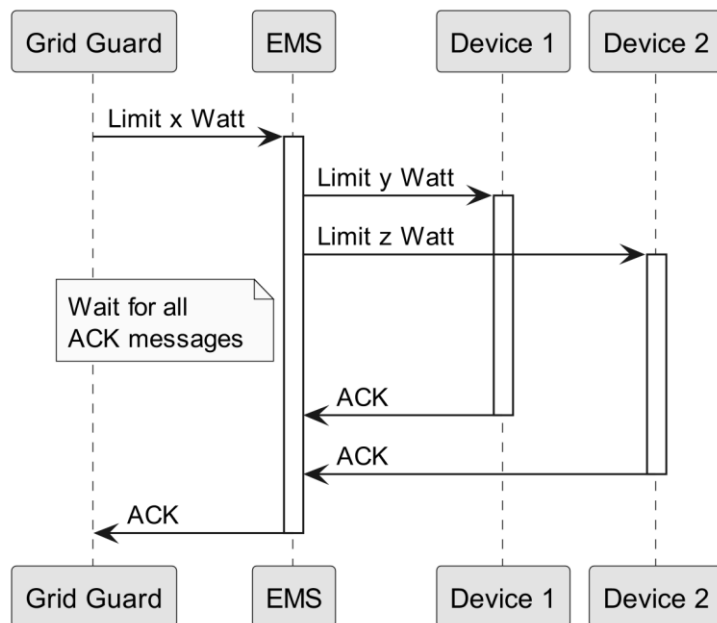


Figure 1: Example of EMS responding to grid's power limitation with full ACK-chain

However, there may be circumstances that prevent the EMS from receiving ACK/NACK messages from all relevant devices in time:

- A connected device's reaction time is too slow<sup>\*3</sup>. Hence, the EMS cannot await the device's response until sending its response to the grid's Energy Guard as this must happen in a certain period of time<sup>\*4</sup>.
- Timing or (temporary) connection problems.

In such cases, the EMS needs to evaluate whether the overall limit can be kept or not by other means (e.g. by measuring the devices' power consumption, the power at the grid connection point, the devices' heartbeat messages, etc.).

Regardless of whether the EMS' response is based on ACK-messages from the devices or on the EMS' analysis of other available information, there might be the case that after the EMS acknowledged the grid's power limitation, the EMS later on determines that the overall limit cannot be kept. In such a case the EMS has to deactivate the grid's power limit and inform the grid's Actor Energy Guard by according means<sup>\*5</sup> (see [LPC-923]).

<sup>\*1</sup>: Which devices are affected may vary depending on the country specific regulations.

<sup>\*2</sup>: Although one of the connected devices that is affected by the grid's power limitation has sent a NACK, the EMS possibly can ensure that the overall limit is kept by limiting another device even more.

<sup>\*3</sup>: This problem may arise when a complex device, consisting of several components with one central unit that manages the components, their power consumption and the LPC-realisation, itself needs some time to evaluate whether the limit can be kept or not.

<sup>\*4</sup>: In case of the SPINE protocol, the default value for the maximum response time is 10 seconds.

\*<sup>5</sup>: In case of SPINE-based implementations this is achieved with a subscription mechanism and proper notify-messages to the Energy Guard.

## 2.5 Energy Guard's reaction on NACK messages

The Use Case specification [LPC] does not described in detail how Actor Energy Guard needs to react on NACK messages received from Actor Controllable System as this is primarily the responsibility of the Energy Guard's business logic. However, this section will provide recommendations on specific aspects.

**Keeping or removing a Controllable System:** In case an Energy Guard receives a NACK message from the Controllable System, the Energy Guard SHOULD NOT remove the Controllable System from its list of devices that are used for its energy management. Instead, the Energy Guard should retry the rejected command (see below).

**Retry after connection establishment:** In [LPC] a case is described where a write command might be rejected due to timing circumstances. Excerpt from [LPC] section 2.2:

*"Note: There may occur a timing problem when both CS and EG reboot almost the same time, but the CS finishes its reboot earlier: It may happen that the EG sends a Heartbeat and shortly thereafter a write on the Active Power Consumption Limit within 120s after the EG completed its reboot, but the CS - due to its earlier completion of the reboot - only recognizes the Heartbeat within its 120s period of state "init", switches then to state "unlimited/autonomous" (according to [LPC-906]), and receives the write afterwards. This results in a rejection of the write command just for timing reasons. Therefore, if the EG receives a reject on the write command within the assumed state "init", the EG may simply send Heartbeat and write command once more to make sure the rejection was caused by a timing problem and not because of a permitted exceptional reason."*

In this case, the Energy Guard SHOULD retry its write command as described.

**Retry or reschedule in other cases:** Apart from situations described above, the Energy Guard may as well retry its write message to the same device, but wait a reasonable time for the first retry (e.g. 1 minute) and extend the time with every retry (e.g. one minute longer for every retry). However, it might also be that an Energy Guard needs to impose stronger power limitations to other devices to compensate the rejections of one device.

## 2.6 Actively control a Controllable System's Active Power Consumption Limit

As soon as Actor Energy Guard is configured to control an Actor Controllable System according to Use Case Limitation of Power Consumption\*<sup>1</sup>, the Energy Guard SHOULD do its best effort to keep Actor Controllable System in the states "limited" and "unlimited/controlled".

\*<sup>1</sup>: For a SPINE-based implementation this means that Actor Energy Guard acquired necessary binding permissions from Actor Controllable System.

Note: In theory, the Use Case Limitation of Power Consumption may work well without explicit communication with Actor Energy Guard as long as there is no real limitation of the power consumption required. However, as [LPC] is an essential Use Case to prevent critical grid conditions, it is necessary to monitor the communication chain in order to detect communication problems in time. Keeping an Actor Controllable System in the states "limited" and "unlimited/controlled" is the best approach to monitor the functional communication between these actors and to prove the functionality towards regulatory authorities.

## 2.7 Behaviour of EMS without grid limit provider

This section extends section 2.6 for the case that an EMS with Actor Energy Guard in "LPC instance 2" has not yet been configured to receive an (external) Active Power Consumption Limit from the grid (DSO). In terms of [LPC] this means that the EMS' role as Actor Controllable System in "LPC instance 1" does not know its responsible Actor Energy Guard so far. Therefore, the EMS cannot take over its responsibility to limit the building's power consumption according to [LPC]. In this situation, an EMS SHOULD behave as follows:

- If the EMS as Actor Energy Guard is already configured to control devices according to [LPC]:
  - The EMS SHOULD keep the devices in the state "unlimited/controlled".
- If the EMS as Actor Energy Guard is not yet configured to control devices according to [LPC]:
  - The EMS SHOULD NOT try to control yet uncontrolled devices according to [LPC]\*<sup>1</sup>,\*<sup>2</sup>.

\*<sup>1</sup>: For a SPINE-based implementation this means that the EMS SHOULD NOT acquire necessary binding permissions from potential Actor Controllable Systems.

\*<sup>2</sup>: An uncontrolled device (Controllable System) will then switch to the state "unlimited/autonomous" as it does not receive heartbeat and other signals from the EMS.

In theory, a building may contain several devices with EMS functionality including [LPC]. In this case, it is advised to first select the finally responsible EMS and configure it to receive limits from the grid. Afterwards, the EMS can then be configured to control devices.

## 2.8 Consideration of internal or remote communication problems of a Controllable System

This section does NOT address communication problems between Actor Energy Guard and Actor Controllable System. Rather, this section addresses the behaviour of a Controllable System for the following case:

- The communication interface between the Energy Guard and the Controllable System is active, but communication problems "within" the Controllable System prevent that commands received from the Energy Guard to limit the power consumption don't lead to a proper power consumption limitation at the power consuming components of the Controllable System.

Example 1: The communication interface of the Controllable System is realised by a hardware component "A" that implements the communication protocol with the Energy Guard. The power consuming component "B" of the Controllable System is a separate component with a separate communication to component "A".

Example 2: Like example 1, but component "A" or "B" requires an additional communication with the vendor's cloud service.

The case described above can be seen as a malfunction of the Controllable System regarding the Use Case [LPC]. Of course, other kinds of malfunctions may as well cause problems including a power consumption beyond the permitted limit. But the key point of the described case is that communication problems "within" a Controllable System are usually easily detectable by the Controllable System. This is esp. relevant as regulation authorities can impose rules on communication paths to

- ensure electricity grid stability even if communication problems with vendor's cloud services arise,
- ensure that users cannot easily bypass electricity grid stability measures by interrupt of easily accessible or configurable communication paths.

It is therefore recommended to detect "internal" communication problems and initiate safety mechanisms like the failsafe state of [LPC] in the present circumstances.

## 2.9 Failsafe Duration Minimum as shared data point

The data point Failsafe Duration Minimum of [LPC] is identical to the Failsafe Duration Minimum of [LPP]. This means both Use Cases SHALL always apply the same value.

If Actor Controllable System supports both Uses Cases [LPC] and [LPP], the Actor SHALL accept an Energy Guard's write command on the Failsafe Duration Minimum if any of the state machines of [LPC] or [LPP] currently permit writing the Failsafe Duration Minimum<sup>\*1</sup>, provided that the Failsafe Duration Minimum is within the permitted value range.

<sup>\*1</sup>: The state machines of [LPC] and [LPP] are independent from each other. [LPC] and [LPP] specify a few conditions when a write on the Failsafe Duration Minimum shall not be evaluated. As long as one of the Use Cases permits writing the Failsafe Duration Minimum, it does not matter if the other Use Case does not evaluate the write command. This also means a write on the Failsafe Duration Minimum SHALL be rejected if neither [LPC] nor [LPP] permit the write command.

## 2.10 Recommendations on change (write) command frequency

Although [LPC] does not impose restrictions on the frequency of updates of the Active Power Consumption Limit (or other data points), it is strongly recommended that an Energy Guard SHOULD NOT write [LPC] data points more often than once every 5 minutes (except for retries due to errors or timing problems).

### 3 Clarifications on the technical part of the Use Case LPC

#### 3.1 Reaction to write commands outside the valid ranges

In case the Controllable System receives a write command from the Energy Guard that is out of the permitted value range<sup>\*1</sup>, the Controllable System SHALL send a NACK response. This means it SHALL NOT discard the message without informing the Energy Guard.

<sup>\*1</sup>: E.g., a write on the Failsafe Duration Minimum ([LPC-022]) with a value higher than 24 hours as defined in [LPC-022/4].

#### 3.2 Usage of Actors and entityTypes

For both Actors (Energy Guard and Controllable System) there is a list of Entity Types that are permitted (see sections 3.2.1.1.1 and 3.2.2.1.1 of the [LPC] UC).

For an EMS that implements both roles of the UC, the Entity Type "CEM" SHALL be used for the Actor Energy Guard and for the Actor Controllable System.

A control box only implements the Actor Energy Guard towards an EMS or end device. For this implementation, the Entity Type "GridGuard" SHOULD be used in order to differentiate it from an EMS in the same network.

#### 3.3 Usage of featureType "Generic" for client functionality

Each Actor SHOULD use a client Feature with featureType "Generic" for all its client functionality of its respective Entity.

Regarding [LPC], an Actor SHALL implement its client and server Features in the same Entity.

Rationale: The technical SPINE solution in [LPC] describes an Actor's server Features (Function tables)<sup>\*1</sup>. With the Use Case discovery information of Actor A, the Entity of its server Features can be detected by the other Actor B<sup>\*2</sup>. But it also means that the Actor's client functionality belongs to this Entity. Having client and server functionality in the same Entity is also essential for Features that require binding. The latter enables a server Feature to verify if a received write command contains a source address of a granted binding partner.

<sup>\*1</sup>: In some Use Cases a given Actor has only client functionality or only server functionality. Use Case discovery returns the Actor's Entity<sup>\*2</sup> regardless of its server or client functionality.

<sup>\*2</sup>: Without going into detail, there are also Use Cases with a dynamic Entity where Use Case discovery can only report a fixed parent Entity. Nevertheless, the occurrence of a well-defined child Entity and its resources determines then the presence of the respective Actor.

### 3.4 Usage of Element measurementId in loadControlLimitDescriptionData

Within the loadControlLimitDescriptionData (as part of the data point Active Power Consumption Limit, [LPC-011]), the Element measurementId is defined. This Element is marked as mandatory in the first column, but in the last column ("Element and value rules") the following is specified:

*"SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId. [...]"*

Despite the description, the "M" in the first column has a higher priority, i.e. the Element is mandatory in any case (not only if the Measurement Feature is available on the same Entity that supports an according measurand). Hence, an Energy Guard SHALL consider the content of loadControlLimitDescriptionData as invalid if the Element measurementId is not set. Hence, if a Controllable System does not provide a corresponding measurand, the Element measurementId SHALL still be set (e.g., to a (relatively high) random number that is not used by any measurand on this Entity).

### 3.5 Binding partner clarifications

The [LPC] UC specifies in section 3.4.1.1: "Only one binding partner is allowed to write the data specified in this Scenario.". This does not mean that only one binding partner is permitted in total for the device. It just means that the Controllable System SHALL NOT permit more than one of its binding partners to modify those datapoints that are changeable by the Energy Guard according to [LPC].

Example: We assume the Controllable System with Use Case [LPC] is just one of several Use Case implemented on a SPINE device called "C". An Energy Guard on device "A" acquires binding permission on the "LoadControl" Feature of device "C". For another Use Case "UCX", device "B" needs as well binding permission on the same "LoadControl" Feature. This means the Feature contains entries for both Use Cases. The entries are distinguishable by their identifiers. In this case, device "C" needs to grant binding permissions to the "LoadControl" Feature for device "A" as well as for device "B". However, it must ensure that device "A" does not alter entries belonging to Use Case "UCX" which is controlled by device "B". Likewise, device "C" needs to prevent that device "B" alters data belonging to Use Case [LPC].

How the Controllable System determines which device to allow changing the [LPC] entries is implementation specific (e.g., the first device that attempts to write an [LPC] entry is permitted, any following device trying to write on the same entry is rejected).

### 3.6 Clarification on permitted value ranges for LPC data points

Section 2.8.1 of [LPC] defines that the data points

- Failsafe Consumption Active Power Limit ([LPC-021])
- Power Consumption Nominal Max ([LPC-041])
- Contractual Consumption Nominal Max ([LPC-042])

423 are always greather than or equal to zero. Even though these value ranges are not explicitly repeated  
424 within the technical part of [LPC], an EG SHALL only write values  $\geq 0W$  for the listed data points. A CS  
425 SHALL react with a NACK to write commands on [LPC-021], [LPC-041] or [LPC-042] with values  $< 0W$ .

426

### 427 **3.7 Heartbeat Elements - presence indication and verification obligations**

428 The heartbeat message contains the following Elements:

- 429 - timestamp
- 430 - heartbeatCounter
- 431 - heartbeatTimeout

432 As a server instance, all three Elements SHALL be set to values as specified in [LPC]. A violation of the  
433 corresponding rules leads to a negative test result within any certification process.

434 A client instance, on the other hand, is not obliged to verify the presence or the correct content of  
435 the Elements heartbeatCounter and heartbeatTimeout (the timestamp Element still needs to be  
436 checked for a correct value).

437 The above mentioned rules guarantee a correct content of the heartbeat messages, but reduce the  
438 workload of a client implementation.

439

## 4 Country specific rules for the Use Case LPC

### 4.1 German EnWG § 14a (2024)

With the release of [BK6-22-300] as part of the EnWG § 14a, the BNetzA (*Bundesnetzagentur für Elektrizität, Gas, Telekommunikation, Post und Eisenbahnen*; Federal Network Agency for Electricity, Gas, Telecommunications, Post and Railway) defined requirements for specific devices in the buildings that need to be limitable by the DSO. The following kind of devices are listed:

- non-public charging stations
- heat pumps (including backup heaters)
- air conditioners
- stationary batteries (inverters).

These devices are called "SteuVEs" (*Steuerbare Verbrauchseinrichtungen*; controllable appliances)

The following sections detail how the Use Case [LPC] can be used to fulfil the new § 14a requirements.

For further details please refer to [BK6-22-300].

Please note that new versions of the law and its additional documents may be released that may change the regulations that are quoted within this document.

#### 4.1.1 Interpretation of LPC limit in Germany

As with the [BK6-22-300] only specific kinds of devices are obliged to follow the DSO's limits, an [LPC] Active Power Consumption Limit does not limit the building's complete consumption at the grid connection point (GCP), but only the consumption of the "SteuVEs".

If more than one "SteuVE" is installed in the building and they are managed by an EMS (this means, they are not individually connected to a control box), the limit that is sent to the EMS limits the combined power consumption of all installed "SteuVEs" (see section 4.1.2, option 2, for details).

#### 4.1.2 Direct vs. EMS controlled limitation

There are two possibilities how devices are limited via [LPC]:

- 1) Directly via the control box
- 2) Via an EMS that receives a limit from the control box and sends proper limits to the end devices

With option 1, the device is connected to the control box and the DSO addresses the device explicitly. If further devices are connected, each receives its own limit. With this option, a building's PV system cannot be used to enable a higher power consumption of the device as it is not taken into account.

With option 2, an EMS receives a limit from the control box. The EMS has to ensure that all SteuVEs together do not consume more power from the grid than permitted by this limit. A local PV system

can be used to permit higher power consumption of the end devices, as the self-produced power reduces the load at the grid connection point. Additionally, the EMS may allocate the power of the different devices individually (based on the customer's needs), as long as the overall limit is kept.

#### 4.1.3 Minimum values for limits

In [BK6-22-300] it is specified that devices are permitted to use a certain amount of power even if a power limitation is active.

It has to be differentiated between the different kind of devices and the amount of devices in the building. For non-public charging stations and stationary batteries different calculations are specified as for heat pumps and air conditioners.

Details can be retrieved from the official appendix no. 1 of the document [BK6-22-300].

#### 4.1.4 SPINE entityType for the SteuVEs

The following table lists the SPINE entityType that SHALL be used by Actor Controllable System on a SteuVE:

| SteuVE device category       | SPINE entityType                |
|------------------------------|---------------------------------|
| non-public charging stations | EVSE                            |
| heat pumps                   | HeatPumpAppliance               |
| air conditioners             | HeatPumpAppliance* <sup>1</sup> |
| stationary batteries         | Inverter                        |

Table 1: SPINE Actors for the SteuVEs

\*<sup>1</sup>: As air conditioners are not part of the [LPC] specification yet, the Actor HeatPumpAppliance SHALL be used instead (as most air conditioners are technically similar to heat pumps).

Additionally, an EMS can be used to receive the DSO's limitations. In this case, the entityType "CEM" SHALL be used by the EMS.

For the control box, the entityType "GridGuard" SHOULD be used in order to differentiate it from an EMS in the same network.

#### 4.1.5 SPINE ACK/NACK messages as provision of proof

The EnWG § 14a requires that the device operator (end customer) can prove the fulfillment of a power limitation. Apart from permitted exceptions and transition rules, this requirement affects each installation, regardless of whether the affected devices and their applications and protocol stacks can already contribute to this proof or not. In case of implementations based upon SHIP/SPINE with the SPINE-based realisation of [LPC], the BNetzA currently considers the requirements for information exchange and verification of grid-oriented control to be already fulfilled (by referring to the [VDE-AR-E 2829-6]).

Background information: Although the high-level part of [LPC] already describes a Controllable System's acceptance (ACK) or rejection (NACK) of Energy Guard commands, these mechanisms are

described in a technology independent way, hence can be realised in various ways. To support a robust signalling of acceptance and rejection of commands, SPINE provides "result" messages to transport "ACK" and "NACK" responses. Additionally, commands and responses are linked together with message numbers in SPINE. In combination with a transport protocol like SHIP, this leads to a reliable response mechanism for the Energy Guard's commands. This mechanism, together with the SPINE realisation of [LPC], fulfils the above-mentioned requirements.

As described in section 2.4, an EMS needs to evaluate whether to send an ACK or NACK back to the grid's Actor Energy Guard. In case the EMS received responses (ACK/NACK) from the affected devices and based its response on them ("full ACK-chain"), the EMS does not need to provide additional logging for the provision of proof to the BNetzA. In contrast, if the EMS' response is based on other information (e.g., the devices' power consumption, the power at the grid connection point, etc.), the EMS must be able to provide this information in case it needs to prove its decision (i.e., it must log this information).

The ACK/NACK messages received at the DSO's Energy Guard (e.g. a control box) via the SPINE protocol are reported as appropriate information to the DSO's backend where the success or failure of the write attempts can be logged for later evaluation and provision of proof to the BNetzA.

#### **4.1.6 Maximum reaction time for power limitation**

The [BK6-22-300] distinguishes between the reaction time of a DSO and the reaction time of a device operator (end customer). Broadly speaking, a DSO must send a power limitation to each affected device operator not later than 5 minutes after the detection of the necessity to limit the power consumption of the respective part of the electricity grid. For the device operator, the [BK6-22-300] requires an "immediate" fulfillment of the received power limitation. However, the requirement to react "immediatly" is mitigated by typical reaction times of a system/devices (i.e. times that can be expounded) as well as necessary delays that would otherwise damage the system/devices. For stationary batteries (inverters) and non-public charging stations, the reaction time to be expected is within seconds, whereas a reaction time of a few minutes can be required by electromechanical devices like heat pumps or air conditioners.